YJ — Premium Wearable AI Voice Recorder

Design Brief — Production Ready

A wearable AI voice recording device. If Apple made a Plaud.ai PIN competitor — rounded rectangle shape, screen on front, microphone holes for directional capture, designed for the neck or collar.

Core Scenario

Worn as a pendant on a cord/chain/lanyard through integrated top loop, or clipped to a shirt collar. One tap to capture. Glance down to confirm. Microphone holes positioned to capture conversation in front of wearer. No phone required for confidence.

Form Factor (Exact Dimensions)

Reference: Plaud PIN = 51 × 21 × 11 mm — rounded rectangle form
YJ Dimensions: 51mm L × 32mm W × 16.5mm H

Dimension	Measurement	Notes
Length	51 mm (2.0")	Same as Plaud PIN — vertical "tallness" maintained
Width	32 mm (1.26")	1.5x Plaud PIN (vs 21mm) — wider, more screen real estate
Thickness	16.5 mm (0.65")	1.5x Plaud PIN (vs 11mm) — accommodates 800-1000mAh battery

Scale: 1.5x total volume of Plaud PIN, wider stance

Shape: Rounded rectangle — like Plaud PIN but wider. Defined straight sides with generously rounded corners (6-8mm radius). NOT a true oval, NOT an egg shape, NOT an organic pebble. Think iPhone with more generous corner curves.

Cross-section: Slightly rounded back and front surfaces for comfort, but the overall silhouette reads as a rounded rectangle from any angle. Edges are softly curved, not sharp.

Weight: 45-60g — substantial quality, comfortable for extended wear, weighted toward bottom half for stable vertical hang

Materials & Finish

Primary Body

- **Material:** Dark titanium oxide (WiFi-conductive)
- **Finish:** Satin-brushed matte

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- **Color:** Deep charcoal gray — sophisticated, not pure black
- **Feel:** Cool, premium heft, fingerprint-resistant, hypoallergenic for skin contact

Front Glass

- **Type:** 2.5D curved edge glass — flat display area for clear viewing, subtle curve at perimeter
- **Integration:** Flows seamlessly into titanium body — no visible step or gap
- **Coverage:** Extends to edges following rounded rectangle shape

Back Surface

- **Finish:** Matching satin titanium — continuous with sides
- **Skin contact:** Smooth, comfortable against bare skin for 8+ hours

Display & Interface — Front Facing

Screen

- **Type:** OLED, portrait orientation
- **Size:** ~1.8" diagonal
- **Coverage:** ~75% of front surface
- **Position:** Centered, with margin for mic holes above and touch buttons below
- **Bezels:** Minimal — titanium oxide bezel (2-3mm) wrapping around edges only

Touch Interface — THREE LABELED BUTTONS

Three capacitive touch buttons on lower portion of OLED screen, arranged vertically:

Button	Label	Icon	Function
Top	**START**	Red solid circle (●)	Begin recording
Middle	**STOP**	Gray square (■)	End recording
Bottom	**SETTINGS**	Gear (⚙)	Device settings

Layout: Vertical stack, centered on lower third of screen

Button style: Each button shows icon + text label in clean sans-serif (Apple SF Pro style)

Size: Minimum 10-12mm tap area each — tappable with one finger while glancing down

Active state: Button glows or inverts color when pressed

Haptic feedback: Subtle vibration on touch

Visibility: Clearly readable at glance when worn at chest level

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Physical Control

- **Location:** Right side edge, upper third — follows the curve
- **Style:** Flush with body surface — nearly invisible, same titanium finish
- **Function:** Power — press-and-hold for on/off, short press for sleep/wake
- **Tactile cue:** Micro-groove texture on button surface — identifiable by touch

LED Indicators

- **Type:** Subtle status lights in screen bezel or edge glow
- **Behavior:** Visible at downward glance, not distracting to others
- **Recording active:** Subtle red pulse at screen edge
- **Battery low:** Amber warning indicator

Audio Capture — Directional Microphone Holes

Mic Array

- **Configuration:** 3-mic beamforming system
- **Direction:** Front-facing capture — focuses on conversations and sound in front of wearer

Microphone Hole Placement

- **Location:** Upper portion of front face, above screen — in the titanium bezel area
- **Pattern:** 6-8 precision-milled holes in a gentle horizontal arc
- **Hole size:** ~1mm diameter each
- **Appearance:** Laser-drilled, titanium-colored interior — subtle, not grille-like
- **Design intent:** Holes appear as intentional design element, not industrial microphone grille
- **Function:** Positioned to capture audio from direction wearer faces (conversations, meetings, ambient sound in front)

Voice Isolation

Function: Beamforming focuses on audio in front of wearer, reduces side/rear noise

Wearability

Neck Attachment

- **Type:** Integrated titanium loop at top of device
- **Style:** Like Plaud PIN — raised loop extension, part of unibody form
- **Protrusion:** 4mm from top edge
- **Opening:** Oval-shaped (8mm × 4mm) — sized for 2-3mm cord/chain/lanyard

- **Integration:** Appears as natural extension of the device form — not a separate ring attached
- **Edges:** Fully rounded — won't snag fabric or irritate skin
- **Stability:** Hangs vertically stable

Clip (Optional)

- **Type:** Magnetic attachment on back
- **Function:** Snaps on for collar wear, removes cleanly for neck wear
- **Positioning:** Holds device vertically, microphone holes facing outward

Balance

- **Weight distribution:** Bottom-heavy — maintains portrait orientation when worn
- **Stability:** Device hangs naturally vertical, doesn't twist or flip

Charging

USB-C Only

- **Location:** Bottom edge, centered
- **Recession:** 0.5mm recessed into edge — protects connector
- **Surround:** Precision-milled titanium, chamfered entry
- **Integration:** Port appears as intentional detail, not afterthought

Charging Status Indicator

- **Location:** Subtle LED ring around USB-C port OR soft screen edge glow
- **While charging:** Gentle amber pulse — breathes slowly, calming rhythm
- **Full charge:** Transitions to steady dim green light
- **Design principle:** "Apple moment" — elegant feedback without visual noise, visible enough to confirm status at a glance, subtle enough to not distract in a dark room

No Wireless Charging

Clean back surface — no charging coil

Branding

- **Name:** YJ
- **Placement:** Laser-etched on back lower portion only
- **Style:** Subtle, catches light at angles
- **Front:** No branding — screen dominates

Scale Reference — CRITICAL

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⚠️ **The device is COMPACT — do not make it appear huge in renders.**

True size context:

- 51mm × 32mm is smaller than a credit card (85.6mm × 53.98mm)
- Similar in size to a large smartwatch face or 3-4 fingers held together
- Thickness (16.5mm) is roughly the width of a USB-C connector

In-hand render requirements:

- Device should rest comfortably on fingers — NOT span the entire palm
- Show device dwarfed by hand — it's a pendant, not a phone
- Include a reference object nearby (coin, pen, AirPods case) for scale accuracy
- The device should look like something you forget is in your pocket, not something bulky

Design Mandate

✓ Oval pebble silhouette — unmistakably soft, organic, continuous curves, no flat planes
✓ Front = screen + microphone holes — OLED glass primary, subtle hole array above for directional audio
✓ Zero visible screws, seams, or joints
✓ Glass flows into titanium — no visible step
✓ Recognizable at distance — "that's the YJ recorder"
✓ Thread loop integrated as natural part of form
✓ Comfortable against skin for 8+ hours
✓ Microphone holes positioned for optimal front-facing capture
✓ Charging status feedback — elegant amber/green indicator

What to AVOID

❌ True oval/egg shape — this is a ROUNDED RECTANGLE with generous corners
❌ Sharp rectangular corners — corners should be generous 6-8mm radius, not tight
❌ Flat sharp edges — edges should be softly curved for comfort
❌ Prominent mic grille that dominates the front — holes should be subtle
❌ Thread loop that looks "bolted on" — must integrate naturally
❌ Bulky thickness that looks like a hockey puck
❌ Screen that looks "stuck on" — glass must flow into body
❌ Plastic-looking finishes
❌ Visible seams between front/back
❌ **Oversized appearance in hand — this is a compact wearable, NOT a phone**
❌ **Tiny or unlabeled buttons — START, STOP, SETTINGS must be clearly readable**
❌ **YJ logo on front or sides — rear only**

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❌ **Bright/distracting charging LEDs — must be subtle and elegant**

Deliverables

1. **Front view** — OLED screen with three labeled touch buttons (● START, ■ STOP, ⚙ SETTINGS), microphone hole arc above
2. **Back view** — Clean surface, subtle YJ laser-etching only
3. **Side/profile view** — Shows thickness, softly curved edges, flush power button on right
4. **Isometric view** — Rounded rectangle form from front-right angle, shows depth and corner radius
5. **Top detail** — Integrated loop close-up, natural extension of form, oval opening
6. **Bottom detail** — USB-C port recessed in edge, precision-milled surround
7. **Mic hole close-up** — Precision pattern (6-8 holes), subtle integration above screen
8. **Wear context — neck** — Device on cord/chain at chest level, vertical orientation
9. **Wear context — collar** — Device clipped to shirt collar, mic facing outward
10. **In-hand render** — CORRECT COMPACT SCALE (51×32mm is smaller than credit card), device rests on fingers
11. **Scale comparison** — Device next to coin (US quarter) for true-size reference
12. **Material swatches** — Dark titanium oxide finish samples
13. **UI close-up** — START, STOP, SETTINGS buttons clearly visible with icons + labels
14. **Charging state render** — Device plugged into USB-C, showing amber pulse indicator (charging) and green steady light (full charge)

Hard Constraints

- ✓ **Exact dimensions:** 51mm × 32mm × 16.5mm — 1.5x Plaud PIN volume
- ✓ **Shape:** Rounded rectangle — defined sides, generous 6-8mm corner radius, NOT a true oval
- ✓ **Front:** OLED screen + microphone holes above — screen dominates
- ✓ **Three labeled touch buttons:** ● START, ■ STOP, ⚙ SETTINGS (vertical stack, icon + text)
- ✓ **Power button** on right side only — flush with edge
- ✓ **Integrated loop** at top — Plaud PIN-style, natural extension, 4mm protrusion
- ✓ **USB-C** on bottom edge — recessed, precision-milled titanium surround
- ✓ **No wireless charging** — clean back surface
- ✓ **YJ branding** rear lower portion only — never on front or sides
- ✓ **Compact scale in renders** — device is 51×32mm, smaller than credit card
- ✓ **Charging status indicator** — Amber pulse while charging, green steady when full
- ✓ **Apple-level minimalism** — refined, distinctive, functional, premium

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Quick Reference Card

Category	Specification
Size	51mm × 32mm × 16.5mm (smaller than credit card)
Shape	Rounded rectangle, 6-8mm corner radius, NOT oval
Material	Dark titanium oxide, satin-brushed
Screen	1.8" OLED, 75% front coverage
Buttons	● START, ■ STOP, ⚙ SETTINGS (icon + text)
Mic	6-8 holes above screen, front-facing
Charging	USB-C only, amber→green status
Loop	4mm protrusion, 8×4mm oval opening
Weight	45-60g, bottom-heavy
Brand	YJ, rear only